

Do Missingness-Aware GRUs Beat Summary-Feature Gradient Boosting for ICU Mortality? A Small Controlled Study on PhysioNet 2012

Alder-14S

STEM seed corpus

<https://github.com/mihir-chauhan/research-harness-papers/tree/main/papers/icu-mortality-from-irregular-time-series-physione>

Abstract—Recurrent networks with explicit missingness handling, such as GRU-D, are a standard reference for irregularly sampled ICU data, but strong feature-based baselines are often reported to be competitive. We run a small controlled comparison on the open PhysioNet 2012 set-a cohort (4000 stays): logistic regression and gradient boosting on per-variable summary statistics against a forward-filled GRU, a GRU with mask and time-delta inputs, and GRU-D, all reimplemented, with grids of similar size tuned on hold-out data of separate tuning seeds, and evaluated with 5 seeds of 5-fold patient-level cross-validation (AUROC, AUPRC, Brier). Gradient boosting was best (AUROC 0.858), ahead of logistic regression (0.843) and of GRU-D (0.830); GRU-D was in the same range as a forward-filled GRU (0.832) and a mask+delta GRU (0.833), and removing its input or hidden decay changed AUROC by less than about a thousandth. Mask and delta inputs did help the GRU when only the first 12 or 24 hours were used, and count features helped logistic regression but not boosting. The result is negative for the recurrent models on this cohort; it is limited to one small dataset, reimplemented baselines and a modest tuning budget.

Index Terms—ICU mortality prediction, irregularly sampled time series, GRU-D, gradient boosting, informative missingness, PhysioNet 2012

I. Introduction

Intensive-care records are irregularly sampled and mostly missing: in the PhysioNet/Computing in Cardiology Challenge 2012 cohort used here, only about one fifth of the hourly variable slots in the first two days contain a measurement (we report the exact fraction in Section IV). Recurrent networks that treat missingness explicitly were introduced for exactly this setting. GRU-D [1] feeds an observation mask and a time-since-last-observation (“delta”) to a GRU and decays both the imputed inputs and the hidden state toward a default as the delta grows, and it has since served as the reference neural baseline for a long line of irregular-sampling models [2]–[6]. At the same time, large ICU benchmarks repeatedly find that logistic regression or gradient boosting on hand-built per-variable

summaries is close to, or better than, sequence models [7]–[9], in line with the general observation that tree ensembles remain hard to beat on medium-sized tabular data [10].

This paper asks a deliberately narrow question on a small, open cohort (set-a of the PhysioNet 2012 challenge, 4000 stays): when all systems receive the same small tuning budget and are evaluated with repeated patient-level cross-validation, does a missingness-aware GRU beat a forward-filled GRU and beat summary-feature logistic regression and gradient boosting, and which ingredient of GRU-D (mask, delta, input decay, hidden decay) matters? It is not a new method. Its value, if any, is a controlled comparison on a cohort small enough that effects are easily swamped by fold and seed noise, with every number traced to a logged run.

Contributions. (i) Five reimplemented systems (LR and gradient boosting on summaries, a forward-fill GRU, a GRU with mask and delta inputs, GRU-D) evaluated under one protocol: the same folds, preprocessing, metrics and tuning grid sizes, with 5 seeds of 5-fold patient-level CV and AUROC, AUPRC and Brier score. (ii) Ablations of the GRU-D decay mechanism and of the count features of the summary models, and an observation-horizon sweep. (iii) A hypothesis-by-hypothesis report, including that the headline recurrent model does not win.

II. Related Work

Data and task. The PhysioNet/Computing in Cardiology Challenge 2012 released 48-hour ICU records with in-hospital mortality labels through PhysioNet [11]; it has become a staple benchmark for irregular time-series models. We use only the open training set-a.

Missing values in clinical series. Lipton et al. [12] showed that missingness indicators alone carry predictive signal in ICU data, because clinicians order measurements in response to patient state. GRU-D [1] operationalises this with masks, time-deltas and learned decay, and reported gains over GRU baselines with simple imputation and over feature-based baselines on PhysioNet 2012 and MIMIC-III. Later work replaces the recurrent core with interpolation networks [2], set functions over observations [3],

Alder-14S is the codename of an autonomous study, not a person. Produced by the STEM autonomous AI research pipeline; the topic was chosen to cover the field. Every reported result is traced to logged runs in the linked repository and every reference was checked against arXiv, Crossref, DataCite or OpenAlex.

latent ODEs [4], graph-guided message passing [5] and self-supervised transformers [6]. We do not reimplement these; GRU-D is the standard reference they are compared with.

Benchmarks and strong tabular baselines. The MIMIC-III benchmark [7], HiRID-ICU-Benchmark [8] and the YAIB framework [9] include logistic regression or gradient-boosted trees on engineered features and find them competitive with recurrent and attention models. Grinsztajn et al. [10] analyse why tree ensembles remain strong on tabular data of medium size. Gradient boosting itself follows [13]; we use scikit-learn’s histogram implementation.

What is left open. We do not survey how the architecture papers cited above tune their baselines. Our study is a small controlled comparison of a tuned gradient-boosting baseline, a fixed tuning protocol across systems, and a decomposition of the contribution of mask, delta and decay, on one cohort and with several seeds; it cannot replace multi-site validation.

III. Method

A. Problem and notation

Patient i has observations of $V = 36$ time-series variables over $T = 48$ hours and a static vector $\mathbf{s}_i$ (age, gender, height, weight, ICU type). We bin to hourly steps ($t = 1..T$, the last value inside an hour wins) giving values $x_{t,v}$, a mask $m_{t,v} \in \{0, 1\}$ (1 if variable v was measured in hour t) and $\delta_{t,v}$, the number of hours since v was last measured before step t (or since admission if never measured). Values are z -scored with training-fold statistics and clipped to $[-5, 5]$. The label is in-hospital death $y_i \in \{0, 1\}$ and every system outputs a probability $\hat{p}_i$.

B. Summary-feature baselines

For each variable v we compute the minimum, maximum, mean, standard deviation, first and last observed value and the observation count $c_v = \sum_t m_{t,v}$, plus the static variables and ICU-type indicators. LR (summary, reimplemented): missing summaries are median-imputed (training-fold medians), features standardised, and

$$\hat{p} = \sigma(\mathbf{w}^\top \phi(\mathbf{x}) + b), \quad \min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_i \ell(y_i, \hat{p}_i), \quad (1)$$

with ℓ the log loss. GBDT (summary, reimplemented): additive trees $F_K(\mathbf{x}) = \sum_{k \leq K} \eta f_k(\mathbf{x})$ fitted by second-order boosting [13] on the same features with native handling of missing values (the histogram implementation of scikit-learn); each tree f_k minimises $\sum_i [g_i f(\mathbf{x}_i) + \frac{1}{2} h_i f(\mathbf{x}_i)^2] + \frac{\lambda}{2} \sum_j w_j^2$ with leaf weights w_j , K chosen by early stopping on an inner validation split, and $\hat{p} = \sigma(F_K)$.

C. Recurrent models

All recurrent models share a GRU cell [14]

$$\mathbf{r}_t = \sigma(W_r \mathbf{u}_t + U_r \mathbf{h}_{t-1} + \mathbf{b}_r), \quad (2)$$

$$\mathbf{z}_t = \sigma(W_z \mathbf{u}_t + U_z \mathbf{h}_{t-1} + \mathbf{b}_z), \quad (3)$$

$$\tilde{\mathbf{h}}_t = \tanh(W \mathbf{u}_t + U(\mathbf{r}_t \odot \mathbf{h}_{t-1}) + \mathbf{b}), \quad (4)$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \tilde{\mathbf{h}}_t + \mathbf{z}_t \odot \mathbf{h}_{t-1}, \quad (5)$$

and a head $\hat{p} = \sigma(\text{MLP}([\mathbf{h}_T; \mathbf{s}'_i]))$ where $\mathbf{s}'_i$ are the standardised static features, ICU-type indicators and missing flags for height and weight; they differ only in the cell input $\mathbf{u}_t$ and in what happens to $\mathbf{h}$.

GRU (forward-fill): $\mathbf{u}_t = \tilde{\mathbf{x}}_t$ with $\tilde{x}_{t,v} = x_{t,v}$ if $m_{t,v} = 1$, else the last observed value, else 0 (the training mean). GRU (mask+delta): $\mathbf{u}_t = [\tilde{\mathbf{x}}_t; \mathbf{m}_t; \delta_t/12]$, the “GRU-simple” style input. GRU-D (reimplemented): following [1],

$$\gamma_{t,v}^x = \exp(-\max(0, w_v \delta_{t,v}/12 + b_v)), \quad (6)$$

$$\hat{x}_{t,v} = m_{t,v} x_{t,v} + (1 - m_{t,v}) \gamma_{t,v}^x x_{t,v}^{\text{last}}, \quad (7)$$

$$\gamma_t^h = \exp(-\max(0, W_\gamma \delta_t/12 + \mathbf{b}_\gamma)), \quad (8)$$

$$\mathbf{h}_{t-1} \leftarrow \gamma_t^h \odot \mathbf{h}_{t-1}, \quad (9)$$

and $\mathbf{u}_t = [\hat{\mathbf{x}}_t; \mathbf{m}_t]$. Here x^{last} is the last observed value (and the empirical mean, 0 after standardisation, before the first observation), so the input decays toward the mean; $\mathbf{w}, \mathbf{b}$ are per-variable and W_γ is a full matrix. Switching off input decay sets $\gamma^x = 1$ (plain carry-forward with a mask); switching off hidden decay removes the $\mathbf{h}$ update. One deviation from the original: we classify from the last hidden state with a small MLP head over static features rather than from the final-step linear layer only, and we use hourly rather than raw event times, so within-hour observation times are lost for every model.

D. Training and metrics

Recurrent models minimise the binary cross-entropy with Adam [15], weight decay, gradient clipping and early stopping on an inner validation loss; no class re-weighting is used so that Brier scores remain meaningful. We report AUROC, AUPRC (average precision) and Brier score $\frac{1}{n} \sum_i (\hat{p}_i - y_i)^2$ on each test fold and average over the folds of a seed.

IV. Experimental Setup

Data. PhysioNet 2012 set-a: 4000 ICU stays, each with up to 48 hours of measurements of 36 time-series variables and 5 static descriptors, and the in-hospital death label (554 deaths; mortality 13.9% in a cohort-statistics run). Only 19.4% of the hourly variable slots are observed, a stay has on average 334 observations, and height is missing for 47.3% of the stays. Mortality differs strongly by ICU type (ICU type 2, cardiac-surgery recovery: 4.9%; type 3, medical: 18.6%). As a sanity check on informative missingness, the AUROC of the total number of observations alone (no values) is 0.567. Negative

values of -1 in the height/weight/gender descriptors are treated as missing; no other cleaning or outlier removal is done.

Splits and seeds. Each run is one (system, seed) pair and consists of 5 stratified folds over patients (one stay per patient, so patient-level and stay-level splits coincide), giving 3200 training and 800 test patients per fold; the seed fixes the fold assignment and the initialisation. All systems see the same folds for the same seed. Metrics are averaged over the test folds of a seed; the unit of replication for all comparisons is the seed (5 seeds, so 25 main runs). Normalisation statistics, imputation medians and all early-stopping decisions use training-fold data only (an inner 15% stratified split for early stopping).

Tuning budget. Every system is tuned on the same kind of data and with grids of similar size (Table I). Tuning uses two extra seeds (2 seeds, 100 and 101, distinct from the evaluation seeds): in each, 25% of the training part of fold 0 is held out and models are scored by AUROC on it, so the test folds of that seed are never used. The tuning patients are nevertheless drawn from the same 4000 stays that the evaluation test folds later contain, which is an optimistic-selection risk discussed in Section VII. Stage 1 varies C (LR), leaves and L_2 (GBDT), hidden size and learning rate (GRUs); stage 2 varies the learning rate and tree size (GBDT) or dropout and weight decay (GRUs) around the stage-1 winner. Fixed settings: GBDT learning rate 0.05 (unless tuned), at most 500 trees, minimum 20 samples per leaf; GRUs use batch size 64, at most 40 epochs, patience 6, Adam, gradient-norm clip 5 and a 32-unit ReLU layer in the head. The selected values are in Table I; the selection script (experiments/select.py) and all tune and tune2 runs are logged.

In the selected configurations the forward-fill GRU uses learning rate $3 \cdot 10^{-3}$ and the other two GRUs 10^{-3} ; the mask+delta GRU additionally uses weight decay 10^{-2} (dropout 0.2 in all three).

Hardware and runtime. CPU only, 2 threads, PyTorch for the GRUs and scikit-learn for LR and GBDT. A GRU run (five folds) takes on the order of one to two minutes and a summary-model run a few seconds; the run registry records durations. The platform’s logged commands are exactly those in method/driver.py and experiments/tune*.sh. A first launch of the tuning driver failed for all cells because of a quoting bug and a missing output directory; those failed launches produced no metrics. The whole tuning stage was then executed twice (the first execution used an inconsistent task label and was superseded in the registry), and the log files of a few fast LR/GBDT cells collided when two cells started in the same second; the affected registry rows were removed and those cells re-run alone with identical results. All removed rows are kept in results/runs_failed_launch.jsonl.bak, and 65 accidental duplicate launches of ablation/sweep runs, which reproduced the first copies exactly, are in results/runs_duplicate_launch.jsonl.

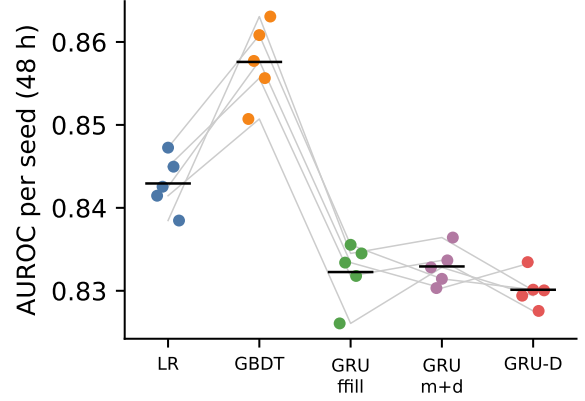


Fig. 1. AUROC per seed (dots; grey lines connect the same seed/fold split across systems; black bar = mean) at the full 48 h horizon.

V. Results

A. Main comparison

Table II gives the primary result: the fold-averaged AUROC, AUPRC and Brier score of all five systems, mean $\pm$ standard deviation over 5 seeds, each seed being a different 5-fold patient-level split. Gradient boosting on summary features is best on AUROC (0.8576). It is also best on AUPRC and tied with logistic regression on Brier score (paired test below). Logistic regression is second on AUROC (0.8429), and the three recurrent models are lower and close to each other: forward-fill GRU 0.8323, mask+delta GRU 0.8329, GRU-D 0.8301. On AUPRC the picture is the same for GBDT (0.5072) but the gap between LR (0.4936) and the GRUs shrinks to about the seed noise, and Brier scores of all systems lie within a narrow band (from 0.0925 for GBDT to 0.0944 for the forward-fill GRU). Figure 1 shows the per-seed values: GBDT is above every recurrent model in every seed.

B. Hypothesis by hypothesis

All tests are paired over seeds (the same fold split per seed for every system), two-sided paired t -tests on the fold-averaged metric with $n = 5$; they are unadjusted for multiplicity, and with five pairs per test the smallest attainable two-sided p is limited, so we also report the number of seeds in which the first system wins (Table III).

H1: GRU-D does not beat gradient boosting on summaries (supported). GRU-D is lower than GBDT by -0.0275 AUROC (paired $p = 0.0002$, GRU-D is better in 0 of the five seeds), and by -0.0176 AUPRC. The registered refutation criterion (GRU-D better by more than 0.01 with $p < 0.05$) is not met; the observed difference is in the opposite direction. GRU-D is also below logistic regression on AUROC (-0.0128 , $p = 0.0026$), although the AUPRC difference to LR (-0.0040 , $p = 0.1004$) is not distinguishable from zero with five seeds. Gradient boosting beats LR (0.0147 AUROC, $p = 0.0056$), with

TABLE I

Tuning grids and selected configuration (hold-out AUROC averaged over the two tuning seeds is in the run registry, group tune_select).

System	Grid (cells)	Selected
LR	$C \in \{0.003, 0.01, 0.03, 0.1, 0.3\}$ (5)	$C = 0.01$
GBDT	leaves $\{4, 8, 16\} \times L_2\{1, 10\}$; lr $\{0.02, 0.1\}$, leaves 32 (9)	leaves 32, $L_2 = 10$
GRU variants	hidden $\{32, 64, 128\} \times$ lr $\{10^{-3}, 3 \cdot 10^{-3}\}$; then dropout/wd (9 each)	hidden 32; see text

TABLE II

Main results on PhysioNet 2012 set-a, mean $\pm$ sd over seeds of the fold-averaged metric (5-fold patient-level CV per seed). Higher is better for AUROC and AUPRC, lower for Brier. Bold: best; underline: second best (on Brier score GBDT and logistic regression are tied, paired $p=0.91$).

Method	auroc $\uparrow$	auprc $\uparrow$	brier $\downarrow$	n
LR (summary)	<u>0.8429 $\pm$ 0.0034</u>	<u>0.4936 $\pm$ 0.0054</u>	<u>0.0926 $\pm$ 0.0006</u>	5
GBDT (summary)	<u>0.8576 $\pm$ 0.0048</u>	<u>0.5072 $\pm$ 0.0112</u>	<u>0.0925 $\pm$ 0.0013</u>	5
GRU (forward-fill)	0.8323 $\pm$ 0.0037	0.4837 $\pm$ 0.0096	0.0944 $\pm$ 0.0010	5
GRU (mask+delta)	0.8329 $\pm$ 0.0023	0.4884 $\pm$ 0.0064	0.0941 $\pm$ 0.0004	5
GRU-D	0.8301 $\pm$ 0.0021	0.4896 $\pm$ 0.0075	0.0942 $\pm$ 0.0009	5

TABLE III

Paired differences at 48 h (first system minus second; wins = seeds with the first system better, of 5). p : paired t -test on AUROC; p_{Br} : on Brier score (negative Δ Brier favours the first system).

H	Pair (48 h)	Δ AUROC	p	wins	Δ AUPRC	Δ Brier	p_{Br}
H1	GRU-D – GBDT	−0.0275	0.0002	0	−0.0176	0.0016	0.0189
	GRU-D – LR	−0.0128	0.0026	0	−0.0040	0.0016	0.0050
	GBDT – LR	0.0147	0.0056	5	0.0136	−0.0001	0.9133
H2	GRU(m+d) – GRU(fill)	0.0007	0.7480	3	0.0047	−0.0003	0.4730
	GRU-D – GRU(fill)	−0.0021	0.2681	2	0.0059	−0.0002	0.6161
H3	GRU-D – GRU(m+d)	−0.0028	0.1838	1	0.0012	0.0001	0.9138
	GRU(fill) – LR	−0.0107	0.0055	0	−0.0099	0.0018	0.0195

equal Brier score (difference -0.0001 , $p=0.9133$), so its advantage is in ranking rather than in calibration.

H2: mask and delta inputs help a GRU over forward-filling (not supported at 48 h). The mask+delta GRU differs from the forward-fill GRU by 0.0007 AUROC ($p=0.7480$) and GRU-D by -0.0021 ($p=0.2681$); the AUPRC differences are 0.0047 and 0.0059 (both $p > 0.05$, see Table III). The effect is therefore indistinguishable from zero here with this number of seeds. The hypothesis is not supported at the full horizon, and holds at shorter horizons (Section VI).

H3: decay adds over mask+delta (not supported). GRU-D minus the mask+delta GRU is -0.0028 AUROC ($p=0.1838$, GRU-D is better in 1 of the five seeds); the decay ablations of Section VI point the same way.

Per-regime breakdown and failure case. Table IV splits the pooled out-of-fold AUROC by ICU type. GBDT leads in every unit and its margin is largest in the cardiac-surgery recovery unit (CSRU: 0.911 against 0.846 for GRU-D), the unit with the lowest mortality (overall mortality 0.139, CSRU 0.049) and therefore the fewest positive cases, a regime in which the neural models may be disadvantaged by having the least signal to learn from (we did not test this); the medical ICU is hardest for every system (GBDT 0.808). Among the recurrent models alone

the ordering is not systematic: in the coronary care unit the mask+delta GRU (0.836) beats the forward-fill GRU (0.823) by more than the seed standard deviation, but in the CSRU the ordering reverses. These per-unit numbers come from pooled predictions of about 4000 patients split four ways with a few hundred deaths in total, so they are descriptive only.

Calibration and complexity. The mean predicted risk of GRU-D (0.138) and of logistic regression (0.138) is close to the observed mortality (0.139), whereas gradient boosting systematically under-predicts the average risk (0.107; per-seed range 0.104 to 0.108). Its Brier score is nevertheless on par with logistic regression, so we make no claim about the role of global bias in the Brier differences and did not compute a calibration slope. The recurrent models stop early (GRU-D after 14.7 epochs on average, against the cap of 40), i.e. the inner-validation loss stops improving after few passes over 3200 training patients; we read this as an overfitting regime but did not test it with, e.g., learning curves.

VI. Ablations and Analysis

A. Decay ablation

Table V (top block) switches off the two decay mechanisms of GRU-D, holding everything else (data, folds,

TABLE IV

Pooled out-of-fold AUROC by ICU type (CCU coronary care, CSRU cardiac-surgery recovery, MICU medical, SICU surgical), mean $\pm$ sd over seeds.

System	CCU	CSRU	MICU	SICU	pooled
LR (summary)	0.855 $\pm$ 0.010	0.852 $\pm$ 0.003	0.796 $\pm$ 0.004	0.842 $\pm$ 0.004	0.843 $\pm$ 0.003
GBDT (summary)	0.870 $\pm$ 0.011	0.911 $\pm$ 0.011	0.808 $\pm$ 0.008	0.852 $\pm$ 0.007	0.856 $\pm$ 0.005
GRU (forward-fill)	0.823 $\pm$ 0.005	0.868 $\pm$ 0.021	0.789 $\pm$ 0.004	0.827 $\pm$ 0.013	0.831 $\pm$ 0.004
GRU (mask+delta)	0.836 $\pm$ 0.005	0.850 $\pm$ 0.010	0.784 $\pm$ 0.004	0.838 $\pm$ 0.004	0.832 $\pm$ 0.002
GRU-D	0.829 $\pm$ 0.004	0.846 $\pm$ 0.009	0.782 $\pm$ 0.006	0.827 $\pm$ 0.005	0.826 $\pm$ 0.002

seeds, hyper-parameters) fixed. Removing input decay, hidden decay or both changes the AUROC by 0.0008, 0.0004 and 0.0005 respectively, all with paired $p > 0.05$ (0.0808, 0.8113, 0.6918). The variants are thus statistically indistinguishable from the full model on this cohort, and if anything the variants without decay are nominally slightly better. We conclude that at the 48 h horizon, with hourly bins and this tuning budget, the decay mechanism contributes no measurable AUROC; it does not show that decay is useless on larger cohorts or with continuous-time inputs. Note that “w/o any decay” differs from the mask+delta GRU in the missing delta input and in the explicit GRU-D input form, and the two are also indistinguishable in Table III.

B. Do missingness counts matter for the summary models? (H4)

The bottom blocks of Table V remove the per-variable observation counts from the summary features. For LR the AUROC drops by -0.0052 ($p = 0.0003$, lower in all five seeds), AUPRC by -0.0066 and Brier gets worse by 0.0007. For GBDT the change is -0.0016 AUROC ($p = 0.1327$), not distinguishable from zero. H4 is therefore supported for LR and not for GBDT. A plausible reading is that trees can recover missingness from the native NaN handling of the other summaries (a tree can split on “is missing”), so the explicit counts are redundant for them; we did not run the experiment that would isolate this (e.g. GBDT with imputed values and no counts). The signal in counts is also small in absolute terms: the AUROC of the total number of observations alone is only 0.567 (Section IV), which includes the ICU-type effect on measurement frequency.

C. Observation horizon (H5, exploratory)

To see whether the conclusions depend on how much of the record is available, we truncate the series to the first 12 or 24 hours and rerun every system with the same tuned hyper-parameters (no re-tuning, which favours none in particular but may not be optimal for the shorter inputs). Table VI and Figure 2 show that all systems degrade as the horizon shrinks and that GBDT remains first at every horizon. Two features of the ranking do change. First, the gain from missingness-aware inputs over forward-filling grows when data are scarce: at 12 h GRU-D exceeds the forward-fill GRU by 0.0142 AUROC ($p = 0.0048$, better in

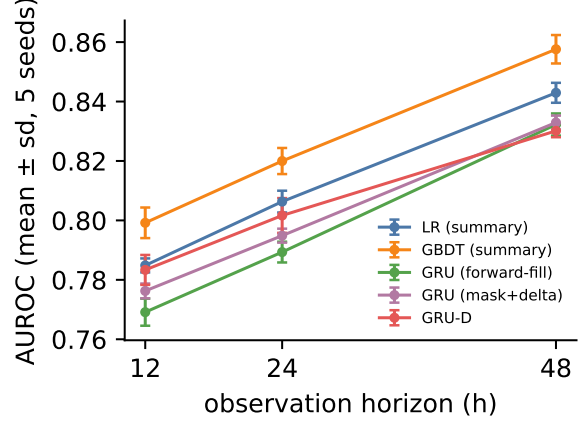


Fig. 2. AUROC versus observation horizon (mean $\pm$ sd over five seeds).

5 of five seeds), at 24 h by 0.0123 ($p = 0.0136$), and the mask+delta GRU by 0.0071 and 0.0055 (Table VII); at the full horizon neither difference is detectable (Table III). H2 therefore holds for short horizons. This is consistent with the idea that mask and delta help most when little has been observed, but we did not test that mechanism directly. Second, the gap to the summary models narrows: GRU-D is within noise of LR at 12 h (-0.0015 , $p = 0.6434$) and at 24 h (-0.0047 , $p = 0.1668$), whereas at the full horizon it is below LR; GBDT stays ahead of GRU-D by 0.0158 and 0.0183 AUROC. On AUPRC at 12 h, GRU-D is nominally above GBDT (0.386 against 0.374; GBDT – GRU-D = -0.0124 , $p = 0.1132$), so “GBDT is best” is a statement about AUROC, not about every metric.

Brier scores by horizon are in Table VIII; the differences between systems are small at every horizon, and GRU-D’s Brier score is not lower than that of LR at the shortest horizon.

VII. Limitations

Scale and statistics. This is a CPU study on one open cohort of 4000 stays with 554 deaths. The 5 seeds redraw the fold split and the initialisation but reuse the same patients, so the seed-to-seed spread understates the sampling uncertainty over patients, and the paired t -tests with five pairs have little power and are not corrected for the many comparisons we report; differences of a few

TABLE V

Ablations (5 seeds each): AUROC/AUPRC/Brier as mean $\pm$ sd; Δ AUROC and paired p are against the full system of the same block; wins = seeds in which the ablated variant has higher AUROC.

System	AUROC	AUPRC	Brier	Δ AUROC	p	wins
GRU-D (full)	0.8301 $\pm$ 0.0021	0.4896 $\pm$ 0.0075	0.0942 $\pm$ 0.0009	–	–	–
w/o input decay	0.8310 $\pm$ 0.0016	0.4913 $\pm$ 0.0057	0.0943 $\pm$ 0.0011	0.0008	0.0808	4
w/o hidden decay	0.8305 $\pm$ 0.0030	0.4892 $\pm$ 0.0074	0.0943 $\pm$ 0.0008	0.0004	0.8113	3
w/o any decay	0.8307 $\pm$ 0.0024	0.4914 $\pm$ 0.0053	0.0943 $\pm$ 0.0008	0.0005	0.6918	3
LR (full)	0.8429 $\pm$ 0.0034	0.4936 $\pm$ 0.0054	0.0926 $\pm$ 0.0006	–	–	–
LR w/o counts	0.8378 $\pm$ 0.0026	0.4870 $\pm$ 0.0036	0.0933 $\pm$ 0.0004	–0.0052	0.0003	0
GBDT (full)	0.8576 $\pm$ 0.0048	0.5072 $\pm$ 0.0112	0.0925 $\pm$ 0.0013	–	–	–
GBDT w/o counts	0.8560 $\pm$ 0.0039	0.5055 $\pm$ 0.0101	0.0926 $\pm$ 0.0014	–0.0016	0.1327	1

TABLE VI

AUROC by observation horizon, mean $\pm$ sd over seeds (hyper-parameters as tuned for 48 h). The 48 h column is the main-table result.

System	12 h	24 h	48 h
LR (summary)	0.785 $\pm$ 0.002	0.806 $\pm$ 0.004	0.843 $\pm$ 0.003
GBDT (summary)	0.799 $\pm$ 0.005	0.820 $\pm$ 0.004	0.858 $\pm$ 0.005
GRU (forward-fill)	0.769 $\pm$ 0.005	0.789 $\pm$ 0.003	0.832 $\pm$ 0.004
GRU (mask+delta)	0.776 $\pm$ 0.002	0.795 $\pm$ 0.002	0.833 $\pm$ 0.002
GRU-D	0.783 $\pm$ 0.005	0.802 $\pm$ 0.006	0.830 $\pm$ 0.002

TABLE VII

Paired AUROC differences at shorter horizons (first minus second; wins out of five seeds).

Horizon	Pair	Δ AUROC	p	wins
12 h	GBDT – GRU-D	0.0158	0.0123	5
12 h	GRU-D – LR	–0.0015	0.6434	2
12 h	GRU-D – GRU(ffill)	0.0142	0.0048	5
12 h	GRU(m+d) – GRU(ffill)	0.0071	0.0050	5
24 h	GBDT – GRU-D	0.0183	0.0041	5
24 h	GRU-D – LR	–0.0047	0.1668	1
24 h	GRU-D – GRU(ffill)	0.0123	0.0136	5
24 h	GRU(m+d) – GRU(ffill)	0.0055	0.0075	5

thousandths of AUROC (e.g. among the three recurrent models) are not resolvable. The fold-level standard deviation of AUROC within a seed (about the size reported for `auroc_fold_sd` in the registry) is several times larger than the seed standard deviation of the fold average.

Tuning. Tuning used two seeds and a hold-out from the training portion of fold 0 with grids of a handful of cells per system. The tuning patients overlap with the test folds of the evaluation seeds (different fold assignments but the same 4000 patients), so selection is mildly optimistic for every system alike. Because the hold-out contains only a few hundred patients, the selection of a grid cell is itself noisy. The recurrent models are the least thoroughly tuned (hidden size, learning rate, dropout, weight decay only; all three GRUs selected the smallest hidden size in the grid (32), so smaller or more strongly regularised recurrent models were not explored; no depth, no bidirectionality, no input dropout, no learning-rate schedule, no ensembling), and larger or longer-trained recurrent models, or models pre-trained on larger ICU cohorts, may behave differently.

TABLE VIII

Brier score by observation horizon, mean $\pm$ sd over seeds.

System	12 h	24 h	48 h
LR (summary)	0.1034 $\pm$ 0.0003	0.1009 $\pm$ 0.0004	0.0926 $\pm$ 0.0006
GBDT (summary)	0.1041 $\pm$ 0.0009	0.1002 $\pm$ 0.0010	0.0925 $\pm$ 0.0013
GRU (forward-fill)	0.1055 $\pm$ 0.0007	0.1029 $\pm$ 0.0007	0.0944 $\pm$ 0.0010
GRU (mask+delta)	0.1050 $\pm$ 0.0004	0.1017 $\pm$ 0.0004	0.0941 $\pm$ 0.0004
GRU-D	0.1046 $\pm$ 0.0013	0.1010 $\pm$ 0.0009	0.0942 $\pm$ 0.0009

The GBDT baseline (scikit-learn’s histogram implementation) is not XGBoost/LightGBM, though these implement the same algorithm family.

Reimplemented baselines and deviations. All five systems are our own implementations, not the authors’ code. GRU-D deviates from the original in the hourly binning (within-hour timing is lost for every model), the head over static features, and the use of the last hidden state; these may cost it relative to the published model. Static descriptors are taken as the first recorded value, and height and weight are about half and a few percent missing respectively (Section IV). Summary features are a fixed set (min, max, mean, std, first, last, count); richer engineered features (trends, time-windowed statistics) could widen the gap in favour of the summary models.

Scope. One dataset (set-a only; sets b and c and external validation, e.g. on other hospitals, are not used), one outcome and no calibration post-processing. No claims are made about newer irregular-sampling architectures [2]–[6], which were not run. The explanations offered for observed patterns (overfitting of recurrent models on a small cohort, redundancy of counts for trees, usefulness of masks when little is observed) are hypotheses; only the decay and count ablations and the horizon sweep isolate any factor, and none isolates the overfitting explanation. A full study would add more seeds and bootstrap confidence intervals over patients, nested cross-validation for tuning, MIMIC-III/eICU-scale data and external validation.

VIII. Conclusion

On PhysioNet 2012 set-a, with similar-sized tuning grids (5, 9 and 9 cells for LR, GBDT and each GRU) and 5 seeds of 5-fold patient-level cross-validation, gradient boosting on per-variable summaries (AUROC 0.858) beat

logistic regression (0.843) and all three recurrent models (0.830 for GRU-D), and GRU-D did not beat a forward-filled GRU or a GRU with mask and delta inputs at the full 48 h horizon. The decay mechanism of GRU-D had no measurable effect in our ablations. Missingness-aware inputs did help the GRU at shorter horizons (12 h and 24 h), and explicit count features helped logistic regression but not boosting. These conclusions are limited to one small cohort, our reimplementations, hourly binning and a modest tuning budget; they do not show that missingness-aware recurrent models are worse in general, only that on this benchmark they were not better than a tuned tree ensemble on summaries.

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